Department of Mechanical and Production Engineering

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Assignment 2: State of Strain

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Definition of The Problem

Let us consider a continuum and a Cartesian reference system $(O; \boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3)$. The following displacement vector field is measured:

$$\begin{aligned} u_1(x_1,x_2,x_3) &= x_1x_3 + x_2^2 + 1, \\ u_2(x_1,x_2,x_3) &= x_1x_3^2 + x_1x_2, \\ u_3(x_1,x_2,x_3) &= x_1^2 + 1, \end{aligned} \tag{1}$$

where (x_1, x_2, x_3) are the Cartesian coordinates of a point within the continuum.

Problem 1

Consider three material points on the continuum

- $\boldsymbol{P}_1$, with coordinates: $\begin{pmatrix} 1 & 0 & 1 \end{pmatrix}$;
- $\boldsymbol{P}_2$, with coordinates: $\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$;
- $\boldsymbol{P}_3$, with coordinates: $\begin{pmatrix} 0 & 2 & 1 \end{pmatrix}$;

Then:

- determine the components of the infinitesimal strain and rotation tensor for all points;
- determine the principal deformations and principal direction of deformations for all points.

Derivation

We start by determining the displacement gradient by finding the first derivatives of the displacement field:

$$\begin{aligned} u_{1,1} &= x_3, & u_{1,2} &= 2x_2, & u_{1,3} &= x_1, \\ u_{2,1} &= x_3^2 + x_2, & u_{2,2} &= x_1, & u_{2,3} &= 2x_1x_3, \\ u_{3,1} &= 2x_1, & u_{3,2} &= 0, & u_{3,3} &= 0. \end{aligned}$$

Meaning

$$\nabla \boldsymbol{u} = [u_{i,j}] = \begin{bmatrix} x_3 & 2x_2 & x_1 \\ x_3^2 + x_2 & x_1 & 2x_1x_3 \\ 2x_1 & 0 & 0 \end{bmatrix}.$$

The infinitesimal strain tensor $\boldsymbol{\epsilon}$ is given by

$$\epsilon_{ij} = \frac{1}{2}\left(u_{i,j} + u_{j,i}\right) \implies \boldsymbol{\epsilon}\frac{1}{2}\left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^\intercal\right)$$

and the rotation tensor $\boldsymbol{\omega}$ is given by

$$\omega_{ij} = \frac{1}{2}\left(u_{i,j} - u_{j,i}\right) \implies \boldsymbol{\omega} = \frac{1}{2}\left(\nabla \boldsymbol{u} - (\nabla \boldsymbol{u})^\intercal\right).$$

So

$$\boldsymbol{\epsilon} = \begin{bmatrix} x_3 & \frac{x_3^2+3x_2}{2} & \frac{3x_1}{2} \\ \frac{x_3^2+3x_2}{2} & x_1 & x_1x_3 \\ \frac{3x_1}{2} & x_1x_3 & 0 \end{bmatrix}$$

$$\boldsymbol{\omega} = \begin{bmatrix} 0 & \frac{x_2-x_3^2}{2} & -\frac{x_1}{2} \\ \frac{x_3^2-x_2}{2} & 0 & x_1x_3 \\ \frac{x_1}{2} & -x_1x_3 & 0 \end{bmatrix}.$$

Evaluating these at all three material points yields:

$$\boldsymbol{\epsilon}(P_1) = \begin{bmatrix} 1 & \frac{1}{3} & \frac{3}{2} \\ \frac{1}{2} & 1 & 1 \\ \frac{3}{2} & 1 & 0 \end{bmatrix} \qquad \boldsymbol{\omega}(P_1) = \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & 0 & 1 \\ \frac{1}{2} & -1 & 0 \end{bmatrix}.$$

The principal strains at P_1 is simply the eigenvalues of $\epsilon(P_1)$ and the principal directions are their corresponding eigenvectors. Solving the eigenvalue problem gives:

$$\det(\boldsymbol{\epsilon}(P_1) - \lambda \boldsymbol{I}) = 0 \rightarrow \begin{pmatrix} 2.62977 & -1.24207 & 0.612303 \\ \{1.11589, 0.955931, 1.\} & \{-0.623383, -0.306997, 1.\} & \{15.1742, -22.1489, 1.\} \end{pmatrix}$$

The directions here are non-normalized and must therefore each be normalized as:

$$\frac{\begin{pmatrix} 1{,}11589 & 0{,}955931 & 1 \end{pmatrix}}{\left|\begin{pmatrix} 1{,}11589 & 0{,}955931 & 1 \end{pmatrix}\right|} = \begin{pmatrix} 0.627835 & 0.537837 & 0.562632 \end{pmatrix}$$
$$\frac{\begin{pmatrix} -0{,}623383 & -0{,}306997 & 1 \end{pmatrix}}{\left|\begin{pmatrix} -0{,}623383 & -0{,}306997 & 1 \end{pmatrix}\right|} = \begin{pmatrix} -0.511924 & -0.252107 & 0.821204 \end{pmatrix}$$
$$\frac{\begin{pmatrix} 15{,}1742 & -22{,}1489 & 1 \end{pmatrix}}{\left|\begin{pmatrix} 15{,}1742 & -22{,}1489 & 1 \end{pmatrix}\right|} = \begin{pmatrix} 0.564792 & -0.824394 & 0.0372205 \end{pmatrix}.$$

Hence the infinitesimal strain and rotation tensor for P_1 are

$$\boldsymbol{\epsilon}(P_1) = \begin{pmatrix} 1 & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & 1 & 1 \\ \frac{3}{2} & 1 & 0 \end{pmatrix}$$
$$\boldsymbol{\omega}(P_1) = \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & 0 & 1 \\ \frac{1}{2} & -1 & 0 \end{bmatrix}.$$

And the principal strains and their directions are:

$$\lambda_1 = 2{,}630 \qquad \boldsymbol{n}_1 = \begin{pmatrix} 0{,}628 \\ 0{,}538 \\ 0{,}563 \end{pmatrix}$$
$$\lambda_2 = -1{,}242 \qquad \boldsymbol{n}_2 = \begin{pmatrix} -0{,}512 \\ -0{,}252 \\ 0{,}821 \end{pmatrix}$$
$$\lambda_3 = 0{,}612 \qquad \boldsymbol{n}_3 = \begin{pmatrix} 0{,}565 \\ -0{,}824 \\ 0{,}037 \end{pmatrix}.$$

For Point P_2 we follow the same procedure, first inserting the point into the strain and rotation tensors found as:

$$\boldsymbol{\epsilon}(P_2) = \begin{bmatrix} 1 & 2 & \frac{3}{2} \\ 2 & 1 & 1 \\ \frac{3}{2} & 1 & 0 \end{bmatrix}$$

$$\boldsymbol{\omega}(P_2) = \begin{bmatrix} 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 \\ \frac{1}{2} & -1 & 0 \end{bmatrix}.$$

Solving the eigenvalue problem yields

$$\det(\epsilon(P_2) - \lambda \boldsymbol{I}) = 0 \to \begin{pmatrix} 3.823 & -1.24542 & -0.577581 \\ \{1.57067, 1.467, 1.\} & \{-1.3131, 0.724229, 1.\} & \{0.242431, -0.941227, 1.\} \end{pmatrix}.$$

Normalizing the eigenvectors gives:

$$\frac{\begin{pmatrix} 1{,}570\,67 & 1{,}467 & 1 \end{pmatrix}}{\left|\begin{pmatrix} 1{,}570\,67 & 1{,}467 & 1 \end{pmatrix}\right|} = \begin{pmatrix} 0.662601 & 0.618867 & 0.421859 \end{pmatrix}$$

$$\frac{\begin{pmatrix} -1{,}3131 & 0{,}724\,229 & 1 \end{pmatrix}}{\left|\begin{pmatrix} 1{,}3131 & 0{,}724\,229 & 1 \end{pmatrix}\right|} = \begin{pmatrix} -0.728518 & 0.401808 & 0.554808 \end{pmatrix}$$

$$\frac{\begin{pmatrix} 0{,}242\,431 & -0{,}941\,227 & 1 \end{pmatrix}}{\left|\begin{pmatrix} 0{,}242\,431 & -0{,}941\,227 & 1 \end{pmatrix}\right|} = \begin{pmatrix} 0.173846 & -0.674948 & 0.717094 \end{pmatrix}.$$

The same is done for point P_3 yielding

$$\boldsymbol{\epsilon}(P_3) = \begin{bmatrix} 1 & \frac{7}{2} & 0 \\ \frac{7}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\boldsymbol{\omega}(P_3) = \begin{bmatrix} 0 & \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Solving the eigenvalue problem and normalizing the vectors gives:

$$\det(\epsilon(P_3) - \lambda \boldsymbol{I}) = 0 \to \begin{pmatrix} 3.823 & -1.24542 & -0.577581 \\ \{1.57067, 1.467, 1.\} & \{-1.3131, 0.724229, 1.\} & \{0.242431, -0.941227, 1.\} \end{pmatrix}.$$

$$\frac{\begin{pmatrix} 1{,}570\,67 & 1{,}467 & 1 \end{pmatrix}}{\left|\begin{pmatrix} 1{,}570\,67 & 1{,}467 & 1 \end{pmatrix}\right|} = \begin{pmatrix} 0.662601 & 0.618867 & 0.421859 \end{pmatrix}$$

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$$\frac{\begin{pmatrix} 0{,}242\,431 & -0{,}941\,227 & 1 \end{pmatrix}}{\left|\begin{pmatrix} 0{,}242\,431 & -0{,}941\,227 & 1 \end{pmatrix}\right|} = \begin{pmatrix} 0.173846 & -0.674948 & 0.717094 \end{pmatrix}.$$

P_1

$$\epsilon(P_1) = \begin{bmatrix} 1 & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & 1 & 1 \\ \frac{3}{2} & 1 & 0 \end{bmatrix} \quad \boldsymbol{\omega}(P_1) = \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & 0 & 1 \\ \frac{1}{2} & -1 & 0 \end{bmatrix}$$

$$\lambda_1 = 2{,}630 \qquad \boldsymbol{n}_1 = \begin{pmatrix} 0{,}628 \\ 0{,}538 \\ 0{,}563 \end{pmatrix}$$

$$\lambda_2 = -1{,}242 \qquad \boldsymbol{n}_2 = \begin{pmatrix} -0{,}512 \\ -0{,}252 \\ 0{,}821 \end{pmatrix}$$

$$\lambda_3 = 0{,}612 \qquad \boldsymbol{n}_3 = \begin{pmatrix} 0{,}565 \\ -0{,}824 \\ 0{,}037 \end{pmatrix}$$

P_2

$$\epsilon(P_2) = \begin{bmatrix} 1 & 2 & \frac{3}{2} \\ 2 & 1 & 1 \\ \frac{3}{2} & 1 & 0 \end{bmatrix} \quad \boldsymbol{\omega}(P_2) = \begin{bmatrix} 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 \\ \frac{1}{2} & -1 & 0 \end{bmatrix}$$

$$\lambda_1 = 3{,}823 \qquad \boldsymbol{n}_1 = \begin{pmatrix} 0{,}662 \\ 0{,}619 \\ 0{,}422 \end{pmatrix}$$

$$\lambda_2 = -1{,}245 \qquad \boldsymbol{n}_2 = \begin{pmatrix} -0{,}729 \\ 0{,}402 \\ 0{,}555 \end{pmatrix}$$

$$\lambda_3 = -0{,}578 \qquad \boldsymbol{n}_3 = \begin{pmatrix} 0{,}174 \\ -0{,}675 \\ 0{,}717 \end{pmatrix}$$

P_3

$$\epsilon(P_3) = \begin{bmatrix} 1 & \frac{7}{2} & 0 \\ \frac{7}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \boldsymbol{\omega}(P_3) = \begin{bmatrix} 0 & \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\lambda_1 = 3{,}823 \qquad \boldsymbol{n}_1 = \begin{pmatrix} 0{,}663 \\ 0{,}619 \\ 0{,}422 \end{pmatrix}$$

$$\lambda_2 = -1{,}245 \qquad \boldsymbol{n}_2 = \begin{pmatrix} -0{,}729 \\ 0{,}402 \\ 0{,}555 \end{pmatrix}$$

$$\lambda_3 = -0{,}578 \qquad \boldsymbol{n}_3 = \begin{pmatrix} 0{,}174 \\ -0{,}675 \\ 0{,}717 \end{pmatrix}$$

RESULT

Problem 2

Determine the mean normal strain and the strain deviator tensors, for the points $\boldsymbol{p}_1, \boldsymbol{p}_2$, and $\boldsymbol{p}_3$.

Derivation

The strain $\boldsymbol{\epsilon}$ can be decomposed to a mean normal strain $\boldsymbol{\epsilon}_M$ and a strain deviator tensor $\boldsymbol{\epsilon}_D$ as

$$\boldsymbol{\epsilon} = \boldsymbol{\epsilon}_M + \boldsymbol{\epsilon}_D.$$

The mean normal strain $\boldsymbol{\epsilon}_M$ is defined as

$$\boldsymbol{\epsilon}_M = \frac{1}{3}(\epsilon_{11} + \epsilon_{22} + \epsilon_{33}) = \frac{1}{3}(x_3 + x_1 + 0) = \frac{x_1 + x_3}{3}.$$

And the deviator is

$$\boldsymbol{\epsilon}_D = \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_M \boldsymbol{I} = \begin{bmatrix} x_3 - \frac{x_1+x_3}{3} & \frac{x_3^2+3x_2}{2} & \frac{3x_1}{2} \\ \frac{x_3^2+3x_2}{2} & x_1 - \frac{x_1+x_3}{3} & x_1x_3 \\ \frac{3x_1}{2} & x_1x_3 & -\frac{x_1+x_3}{3} \end{bmatrix}.$$

At P_1 we get

$$\boldsymbol{\epsilon}(P_1) = \begin{bmatrix} 1 & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & 1 & 1 \\ \frac{3}{2} & 1 & 0 \end{bmatrix}$$

$$\boldsymbol{\epsilon}_M(P_1) = \frac{1}{3}(1+1+0) = \frac{2}{3}$$

$$\boldsymbol{\epsilon}_D(P_1) = \boldsymbol{\epsilon}(P_1) - \frac{2}{3}\boldsymbol{I} = \begin{bmatrix} \frac{1}{3} & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & \frac{1}{3} & 1 \\ \frac{3}{2} & 1 & -\frac{2}{3} \end{bmatrix}.$$

At P_2 we get

$$\boldsymbol{\epsilon}(P_2) = \begin{bmatrix} 1 & 2 & \frac{3}{2} \\ 2 & 1 & 1 \\ \frac{3}{2} & 1 & 0 \end{bmatrix}$$

$$\boldsymbol{\epsilon}_M(P_2) = \frac{1}{3}(1+1+0) = \frac{2}{3}$$

$$\boldsymbol{\epsilon}_D(P_2) = \boldsymbol{\epsilon}(P_2) - \frac{2}{3}\boldsymbol{I} = \begin{bmatrix} \frac{1}{3} & 2 & \frac{3}{2} \\ 2 & \frac{1}{3} & 1 \\ \frac{3}{2} & 1 & -\frac{2}{3} \end{bmatrix}.$$

At P_3 we get

$$\boldsymbol{\epsilon}(P_3) = \begin{bmatrix} 1 & \frac{7}{2} & 0 \\ \frac{7}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\boldsymbol{\epsilon}_M(P_3) = \frac{1}{3}(1) = \frac{1}{3}$$

$$\boldsymbol{\epsilon}_D(P_3) = \boldsymbol{\epsilon}(P_3) - \frac{1}{3}\boldsymbol{I} = \begin{bmatrix} \frac{2}{3} & \frac{7}{2} & 0 \\ \frac{7}{2} & -\frac{1}{3} & 0 \\ 0 & 0 & -\frac{1}{3} \end{bmatrix}.$$

P_1

$$\epsilon_M(P_1) = \frac{2}{3}$$

$$\epsilon_D(P_1) = \begin{bmatrix} \frac{1}{3} & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & \frac{1}{3} & 1 \\ \frac{3}{2} & 1 & -\frac{2}{3} \end{bmatrix}$$

P_2

$$\epsilon_M(P_2) = \frac{2}{3}$$

$$\epsilon_D(P_2) = \begin{bmatrix} \frac{1}{3} & 2 & \frac{3}{2} \\ 2 & \frac{1}{3} & 1 \\ \frac{3}{2} & 1 & -\frac{2}{3} \end{bmatrix}$$

P_3

$$\epsilon_M(P_3) = \frac{1}{3}$$

$$\epsilon_D(P_3) = \begin{bmatrix} \frac{2}{3} & \frac{7}{2} & 0 \\ \frac{7}{2} & -\frac{1}{3} & 0 \\ 0 & 0 & -\frac{1}{3} \end{bmatrix}$$

RESULT

Problem 3

Discuss whether the small gradient approximation is satisfied for the given displacement field. Can the infinitesimal deformation strain be used? Hint: Consider evaluating the discarded nonlinear (quadratic) term leading to the infinitesimal strain tensor.

Derivation

The Lagrangian strain is

$$\epsilon^L_{ij} = \frac{1}{2}\left(u_{i,j} + u_{j,i} + u_{k,i}u_{k,j}\right)$$

and the infinitesimal strain is

$$\epsilon_{ij} = \frac{1}{2}\left(u_{i,j} + u_{j,i}\right)$$

which means the discarded non-linear term is

$$\Delta = \frac{1}{2}u_{k,i}u_{k,j} = \frac{1}{2}(\nabla \boldsymbol{u})^\top \nabla \boldsymbol{u}.$$

Evaluating this at P_1 we obtain

$$\Delta(P_1) = \begin{bmatrix} 3 & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \\ \frac{3}{2} & 1 & \frac{5}{2} \end{bmatrix}.$$

Comparing this with the infinitesimal strain at P_1

$$\boldsymbol{\epsilon}(P_1) = \begin{bmatrix} 1 & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & 1 & 1 \\ \frac{3}{2} & 1 & 0 \end{bmatrix}$$

we see that the discarded quadratic term $\Delta(P_1)$ is not small compared to the infinitesimal strain $\boldsymbol{\epsilon}(P_1)$ meaning the small gradient approximation is *not* satisfied at P_1.

At P_2 we get

$$\Delta(P_2) = \begin{bmatrix} \frac{9}{2} & 2 & \frac{5}{2} \\ 2 & \frac{5}{2} & 2 \\ \frac{5}{2} & 2 & \frac{5}{2} \end{bmatrix}$$

compared to the infinitesimal strain at P_2

$$\boldsymbol{\epsilon}(P_2) = \begin{bmatrix} 1 & 2 & \frac{3}{2} \\ 2 & 1 & 1 \\ \frac{3}{2} & 1 & 0 \end{bmatrix}$$

the discarded quadratic term is not small compared to the infinitesimal strain meaning the small gradient approximation is, once again, *not* applicable at P_2.

At P_3 we get

$$\Delta(P_3) = \begin{bmatrix} 5 & 2 & 0 \\ 2 & 8 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

which is several times larger than the infinitesimal strain at P_3

$$\epsilon(P_3) = \begin{bmatrix} 1 & \frac{7}{2} & 0 \\ \frac{7}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

meaning the small gradient approximation is *not* satisfied at P_3 either.

$\boxed{\text{The small gradient approximation is not satisfied for } P_1, P_2 \text{ or } P_3}$ RESULT